Journal of the Energy Institute 113 (2024) 101556

Contents lists available at ScienceDirect

Journal of the Energy Institute

journal homepage: www.elsevier.com/locate/joei

Enhancing light fuel production through catalytic pyrolysis of municipal mixed plastic waste over activated spent FCC catalyst

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ARTICLE INFO

Handling Editor: Dr. Paul Williams

Keywords:
Waste plastics
Catalytic pyrolysis
Activated spent FCC catalyst
Light fuel

ABSTRACT

To address the issue of environmental pollution caused by waste plastics and increase their recycling utilization, a cost-effective chemical recycling method was employed in this study. It involved the use of spent fluid catalytic cracking (FCC) catalysts to convert waste plastics into valuable products. Sequential nitric acid treatment and pore expansion by ammonium hexafluorosilicate methods were employed to activate the spent FCC catalyst, with the aim of improving its physicochemical properties and enhancing the production of light fuel. The results demonstrated that activation treatment of the spent FCC catalyst improved the external surface area and meso-macroporous volumes of the activated catalysts. This process facilitated the initial cracking reaction of polyolefin macromolecules and alleviated the diffusion limitations of reactants and products within the pores of the catalyst, resulting in an increase in the yield of light fuels. Compared to the spent FCC catalyst, the activated catalysts exhibited 5.79–19.49% higher selectivities for light olefins and 21.67–29.76% higher yields for light fuels. These findings provide potential opportunities for the valuable utilization of waste plastics.

1. Introduction

With the increasing consumption of fossil fuels worldwide, the importance of carbon efficiency in energy and economic development has increased [1]. In recent years, enormous amounts of waste plastics have been generated globally, resulting in severe environmental pollution due to the difficulty of degradation. To address these issues and enhance carbon efficiency, there is an urgent need to develop cost-effective recycling methods for waste plastics and achieve their high-value utilization [2].

Mechanical recycling and incineration are commonly utilized for recycling of waste plastics. Mechanical recycling is simpler but has a limited range of applications. Incineration has a wide range of applications and allows the recovery of energy, but it causes serious pollution to the atmosphere. Chemical recycling, which converts waste plastics into light fuels or chemicals, has recently attracted increasing attention [3]. Catalytic pyrolysis is an important method for the chemical recycling of waste plastics, which is highly compatible with waste plastics and provides a high recycling rate [4]. The use of catalysts in pyrolysis reduces the energy barrier for the cracking reaction, enabling the conversion of waste plastics at lower temperatures and improving the selectivity for value products [5]. Solid acid catalysts are commonly used for catalytic pyrolysis, and their structural properties significantly impact the pyrolysis efficiency [6]. Many studies have been devoted to the conversion of different types of plastics (low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), and their mixtures) into valuable light fuels (gasoline and diesel) with different catalysts [7–11]. The resulting light fuels can be used as feedstocks for refineries or fuel for diesel engine generators [12,13]. Miskolczi conducted catalytic pyrolysis experiments on HDPE and PP with ZSM-5 zeolite, and light fuels were produced with yields ranging from 58 wt% to 78 wt%; the light fuel was mainly gasoline fractions [14]. Similarly, Hwang investigated the catalytic pyrolysis of the wax derived from the pyrolysis of mixed waste plastics (LDPE/HDPE/PP/PS) with HY zeolite. The highest yield obtained was 66.90 wt% for the light fuel; the gasoline fraction primarily

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https://doi.org/10.1016/j.joei.2024.101556
Received 3 November 2023; Received in revised form 1 February 2024; Accepted 6 February 2024
Available online 6 February 2024

consisted of *i*-paraffins and the diesel fraction mainly contained olefins [15]. Huang also conducted catalytic pyrolysis on mixed plastics waste (LDPE/HDPE/PP/PS) with MCM-41 zeolite in a fluidized bed, resulting in 65.40 wt% light fuel [11]. Furthermore, Lin explored the catalytic pyrolysis of mixed plastics waste (LDPE/HDPE/PP/PS/PVC) with HUSY zeolite in a fluidized bed and obtained a 51.70 wt% light fuel yield [16]. Considering the high costs of these catalysts, there is a need to develop cost-effective and efficient catalysts that can be used as alternatives in practical applications.

Spent FCC catalysts are potential catalysts for the chemical recycling of waste plastics [17]. The FCC process generates a significant amount of spent catalyst that retains activity for other processes. This makes the use of zero-value spent FCC catalysts economically beneficial in the catalytic pyrolysis of waste plastics. Therefore, many studies have focused on converting waste plastics into valuable light fuel using spent FCC catalysts. Lovás employed the standard microactivity test (MAT) to conduct catalytic pyrolysis of a feedstock comprising wax from HDPE pyrolysis and atmospheric gas oil over spent FCC catalyst, resulting in a 45.20 wt% yield of light fuel [18]. Passamonti performed catalytic pyrolysis of LDPE and vacuum gas oil (VGO) mixtures in a riser simulator reactor, achieving a light fuel yield of approximately 57.50 wt% [19]. Rodríguez conducted catalytic pyrolysis of HDPE and VGO mixtures with spent FCC catalyst in a riser simulator reactor, resulting in a light fuel yield of 56.20 wt% [20]. These studies offer valuable insights into the potential commercial application for chemical recycling methods of waste plastics. However, the spent FCC catalyst undergoes deposition of metallic contaminants and pore blockage, leading to severe deactivation during catalytic cracking [21]. Therefore, activation treatments of the spent FCC catalysts are essential.

This study aimed to enhance the production of light fuel by activating the spent FCC catalyst through sequential treatment with nitric acid and ammonium hexafluorosilicate (AHFS), followed by catalytic pyrolysis of municipal mixed plastic waste (MMPW) in a confined fluidized bed. The acid treatment removed some part of the metallic containments on the spent FCC catalyst and partially restored its pore structure. Due to the large molecular sizes of waste plastics and the tendency to agglomerate, significant steric-hindrance effects are generated. The acid-treatment catalyst was dealuminated using AHFS to increase the pore size. This process aimed to improve the pore connectivity and accessibility of the active sites. Considering the formation of a silica layer after AHFS treatment, the effect of alkali treatment to remove the silica layer on the catalytic performance of the catalysts and the yield of light fuels was investigated.

2. Materials and methods

2.1. Materials

The raw material, referred to as H-MMPW, used in this study was provided by Fuhai Lantian (Beijing) Environmental Protection Technology Company. H-MMPW was obtained through heating and liquefaction of a mixture of heavy oil and municipal mixed plastic waste. Waste plastics (PE, PP, PS, and PVC) accounted for 25 wt% of the H-MMPW. However, the H-MMPW was in a solid-liquid heterogeneous phase and exhibited high viscosity, which did not meet the requirements of the feeding system of the experimental equipment. To address this issue, straight-run diesel was used as a solvent [22,23]. The H-MMPW was mixed with the straight-run diesel at a 1:1 mass ratio at 150 °C to prepare the feedstock, named DH-MMPW. At this temperature, the feedstock was partially dissolved in the straight-run diesel, leading to a significant reduction in viscosity and meeting the requirements of the feed system. The straight-run diesel and spent FCC catalyst used in the experiments were provided by Sinopec Yanshan Petrochemical Company. The physical properties of straight-run diesel, H-MMPW, and DH-MMPW and the metallic element contents of the spent FCC catalyst are listed in Tables 1 and 2, respectively.

Table 1
Properties of the feedstock.

Property	Straight-run diesel	H-MMPW	DH-MMPW
Density at 25 °C (g/cm^3)	0.8592	0.9206	0.8914
Dynamic viscosity at 110 °C (mPa s)	3.53	2897.70	143.87
Distillation (°C)			
IBP-180	3.2	–	0.8
180–350	96.8	4.7	50.3
> 350	–	95.3	48.9
Metallic element content (μg/g)			
Ca	–	2579.59	1138.38
Fe	–	1288.75	430.37
Si	–	12.17	4.89

Table 2
Metallic elements in the spent FCC catalyst.

Metallic element	Content (μg/g)
Ca	4894.37
Cu	70.42
Fe	3110.33
K	856.81
Mg	927.23
Na	892.02
Ni	762.91
V	2582.16

2.2. Catalyst activation treatment

2.2.1. Acid treatment

The spent FCC catalyst was calcined in a muffle furnace at 550 °C for 5 h to remove the residual coke on the catalyst. Subsequently, the catalyst, with particle sizes ranging from 48 to 150 μm, was sieved. The sieved spent FCC catalyst was then subjected to a reaction with a 13 wt% nitric acid solution at a mass ratio of 1:5 under the conditions of 50 °C and 400 rpm for 1 h. After the reaction, the catalyst was filtered, washed to neutrality, and dried at 110 °C for 12 h.

2.2.2. Pore expansion treatment

The acid-treated catalyst was treated for pore expansion as follows. Initially, 20 g of the acid-treated catalyst was introduced into a 50 mL buffer solution containing 0.4 mol/L ammonium acetate. The pH of the buffer solution was adjusted to 6 using a 0.5 mol/L oxalic acid solution. Subsequently, a 50 mL of AHFS solution was added to the slurry at a flow rate of 100 mL/h and reacted at 90 °C for 1 h. Following the reaction, the catalyst was filtered, washed with 2 L of hot distilled water, and dried at 110 °C for 12 h. The resulting catalyst was denoted as A(X)FS, where X represented the mass concentration of the AHFS solution.

Considering the formation of the silica layer on the catalyst surface after AHFS treatment, the study was carried out by treating A(10)FS with 0.1, 0.3, and 0.5 mol/L NaOH solution at 70 °C for 1 h, respectively. After the treatment, the catalyst was filtered, washed to neutrality, and dried at 110 °C for 12 h. The prepared catalyst was named A(X)FS-(Z)B, where Z represented the molar concentration of the NaOH solution.

2.3. Catalyst characterization

An Agilent-5110 inductively coupled plasma emission spectrometer (ICP-OES) was used to determine the metal content in the catalyst. A 20 mg sample of catalyst was dissolved in a solution of HNO_3 and HCl with a volume ratio of 1:3. The resulting solution was then transferred to a volumetric flask for analysis.

N_2 physisorption measurement was performed using a Micromeritics Tristar II 3020 instrument at liquid nitrogen temperature. All samples were heated to 350 °C under vacuum for 6 h. The specific surface area

(S_{bet}) was estimated using the BET method, while the micropore surface area (S_{mic}), external surface area (S_{ext}), and micropore volume (V_{mic}) were calculated using the *t*-plot method. The meso-macroporous volume ($V_{mes\text{-}mac}$) was calculated from the difference between total pore volume (V_{tot}) and microporous volume. The pore size distribution was calculated by applying the Barrett-Joyner-Halenda (BJH) model to the isothermal adsorption branch.

Scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDS) was used for quantitative analysis of the elemental distribution on the catalyst surface. The catalyst was coated with conductive adhesive and then gold-plated to enhance conductivity. Back-scattered electron (BSE) images were obtained using a Hitachi 3400S scanning electron microscopy at 15-25 kV, and energy dispersive spectroscopy results were collected using a Bruker Quantax EDS system at 25 kV.

Pyridine Infrared Spectroscopy (Py-IR) was employed to characterize the acidity properties of the samples using a Tensor 27 pyridine infrared spectrometer. The samples were sealed in the in-situ cell of the IR spectrometer by pressing self-supporting slices. Subsequently, the samples were subjected to a vacuum of 10^{-3} Pa at 350 °C for 2 h to desorb gas molecules from the sample surface and then cooled to room temperature. Pyridine vapor was introduced into the in-situ cell, heated to 200 °C and 350 °C, vacuumed to 10^{-3} Pa again, and held for 0.5 h. The Py-IR spectra were recorded in the wave number range of 400–4000 cm^{-1} for analysis.

2.4. Experimental equipment and procedure

The experiments on catalytic pyrolysis were conducted in a confined fluidized bed reactor, as depicted in Fig. 1. The catalyst was vigorously circulated within the reactor to prevent catalyst de-fluidization caused by the high-viscosity waste plastic. This process ensured rapid and uniform coverage of the waste plastic on the catalyst surface, thereby enhancing heat and mass transfer between the liquefied waste plastic and the catalyst. The reactor consisted of five parts, including the feed system, water atomizing system, reacting system, product separation and collection system, and flue gas collection system. Prior to initiating the reaction, 200 g of the catalyst was loaded into the reactor, and air was used to fluidize the catalyst in the bed. When the vaporizer temperature reached 300 °C, the fluidized carrier gas was switched from air to steam. Upon reaching a reactor temperature of 500 °C, the feedstock in the preheating box was fed into the reactor, and the generated product was subsequently carried out by steam. In this experiment, the temperatures of the preheating furnace and vaporizer were set at 200 °C and 350 °C, respectively. The mass ratio of catalyst to feedstock was 4:1, and the mass space velocity was 9 h^{-1}. The product was cooled using a three-stage condensing system, which separated the product vapor into liquid and gaseous products. After the reaction, steam stripping was conducted for 0.5 h, after which the catalyst was regenerated at 620 °C. The coke content of the catalyst was determined by analyzing the composition of the regenerated flue gas using an infrared gas analyzer.

The product yield from the catalytic pyrolysis of DH-MMPW and straight-run diesel was calculated using equation (1). The theoretical yield of the H-MMPW catalytic pyrolysis products was calculated using equation (2) without considering the interactions between the straight-run diesel and the H-MMPW during the pyrolysis process.

$$Y_{j,i} = \frac{m_{j,i}}{m_{j,total}} \times 100\% \quad (1)$$

$$Y_{H\text{-}MMPW,i} = \frac{m_{DH\text{-}MMPW,i} - m_{Straight\text{-}run\ diesel,i} \times 50\%}{m_{DH\text{-}MMPW,totlal} \times 50\%} \times 100\% \quad (2)$$

where $Y_{j,i}$ was the yield of product *i* from catalytic pyrolysis of *j*, $m_{j,i}$ was the mass of product *i* from catalytic pyrolysis of *j*, and $m_{j,total}$ was the total mass of the *j* feedstock. Here, *j* was the feedstock, and *i* was the gas, liquid, coke, and light fuel product, respectively.

2.5. Product analysis

The composition of the gaseous product was analyzed using an Agilent 7890A gas chromatograph equipped with an FID and a TCD detector. The composition of the liquid product was analyzed using an Agilent 7890A gas chromatograph equipped with an FID detector and a semi-capillary column for simulated distillation. In this analysis, the gasoline fraction ranged from the initial boiling point (IBP)-180 °C, the diesel fraction ranged from 180 to 350 °C, and the heavy oil fraction had a boiling point above 350 °C.

3. Results and discussion

3.1. Catalytic characterization

The textural properties of the catalysts are listed in Table 3. The S-

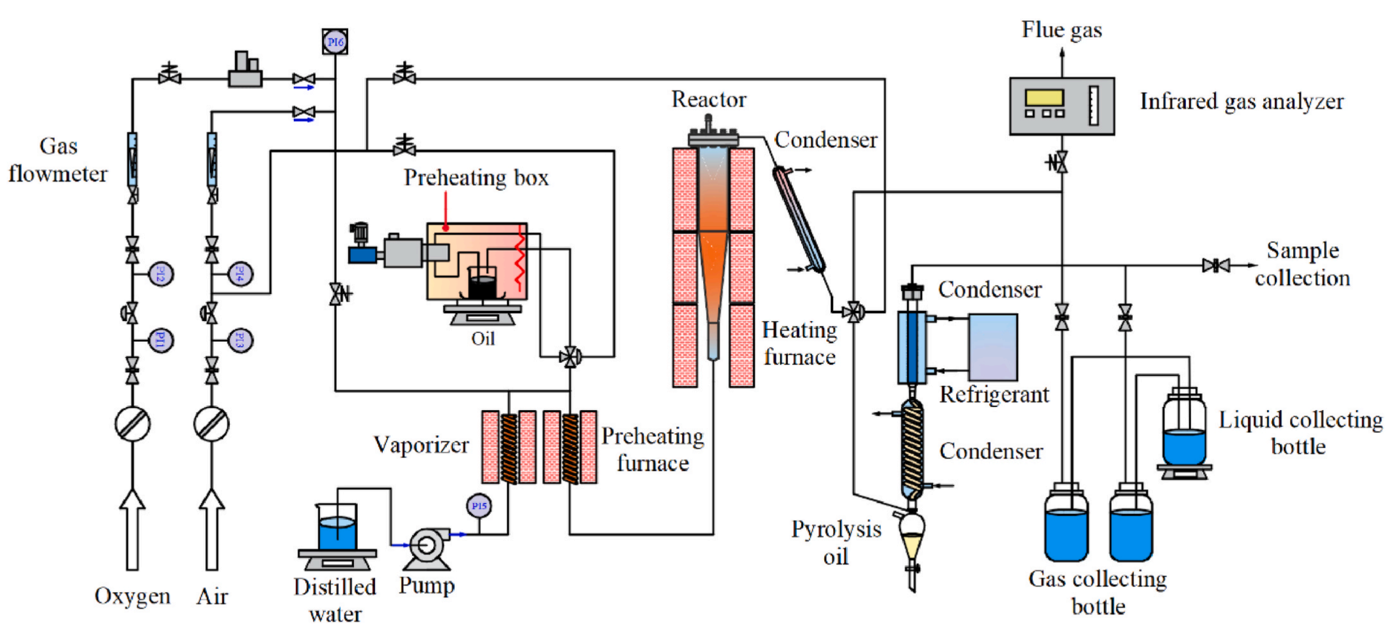

Fig. 1. Schematic diagram of the confined fluidized bed experimental equipment.

Table 3
Textural properties of spent FCC catalyst and activated catalysts.

Sample	S_{bet} (m^2/g)	S_{mic} (m^2/g)	S_{ext} (m^2/g)	V_{total} (cm^3/g)	V_{mic} (cm^3/g)	$V_{mes\text{-}mac}$ (cm^3/g)
S-FCC	92.34	55.05	37.29	0.134	0.022	0.112
A(5)FS	87.69	39.72	47.97	0.141	0.019	0.122
A(10)FS	94.17	30.04	64.13	0.150	0.014	0.136
A(15)FS	79.19	32.44	46.75	0.138	0.014	0.124
A(10)FS-0.1B	68.68	26.59	42.09	0.132	0.010	0.122
A(10)FS-0.3B	41.25	10.77	30.48	0.125	0.005	0.120
A(10)FS-0.5B	29.39	7.07	22.32	0.096	0.003	0.093

FCC catalyst exhibited a higher proportion of microporous structure compared to the activated catalysts, resulting in the highest microporous surface area and microporous volume. After treating the S-FCC catalyst with sequential nitric acid and AHFS, the metallic contaminants on the surface of the catalyst were removed, restoring some of the pores and leading to an increase in pore volume. Moreover, nitric acid and AHFS disrupted the Si–O–Al bonds in the catalyst framework, causing the removal of Al from the framework and enlarging the pore size of the catalyst [24,25]. As shown in Table 3, the microporous surface area and microporous volume decreased for A(5)FS, A(10)FS, and A(15)FS, while the external surface area and meso-macroporous volume increased. This suggested that some of the micropores in the catalysts were transformed into meso- or macropores, confirming that the pore sizes of the catalysts were expanding. The concentration of AHFS influenced the specific surface area and total pore volume of the catalysts, initially increasing and then decreasing. This result was primarily influenced by the interaction between AHFS and the framework Al in the catalysts. When AHFS interacted with the framework Al, a homocrystalline replacement reaction occurred. The Al species departed from the framework to create tetrahedral cavities, while the Si species embedded into the defective sites resulting from de-alumination [26]. However, when the concentration of AHFS exceeded a certain threshold, the embedding rate of Si became lower than the removal rate of Al [27]. Consequently, defective sites formed within the catalyst framework, leading to an expansion in pore size and an increase in pore volume. A(15)FS exhibited a reduction in external surface area and meso-macroporous volume compared to A(10)FS. This decrease could be attributed to some pores being blocked by Si species that were not backfilled into the cavities during the dealumination process [27,28]. To some extent, these Si species as well as the silica layer deposited on the outer surface of the catalyst, might have impeded the penetration of N_2 molecules into the catalyst's pores, thereby contributing to the observed results. This phenomenon has also been reported in the studies conducted by Cruz [29] and Li [28].

To mitigate the impact of the silica layer on the textural properties of the catalysts, the A(10)FS catalyst underwent treatment with NaOH solutions of varying concentrations. The results revealed that A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B exhibited a reduction in specific surface area and pore volume. This reduction can be attributed to the alkali, which not only removed the silica layer on the catalyst's surface but also the framework silica and amorphous silica within the catalyst [30]. Following the sequential dealumination treatment of the spent FCC catalysts, the pores of the catalysts enlarged, and their connectivity improved. This facilitated the diffusion of OH^- inside the catalysts' pores, further disrupting the microporous structure of the catalysts. With an increase in alkali concentration, there was a decrease in the microporous surface area and microporous volume. Simultaneously, the structural stability of the catalyst decreased, leading to a partial collapse of the catalyst, which resulted in a decline in specific surface area and pore volume.

The elemental contents on the surface of the S-FCC catalyst and the activated catalyst were analyzed using SEM-EDS. The results are shown in Figs. 2 and 3, respectively. Fig. 2 revealed the presence of metallic contaminants, including Ni, Fe, and V, on the surface of the S-FCC catalyst. These contaminants could form nodules on the outer surface of the catalyst, either individually or by interaction with Si or Al in the catalyst, through the vitrification process [31,32]. Consequently, the pore entrance became blocked, leading to a decrease in the accessibility of the catalyst's active sites. Fig. 3(a), (b), and (c) demonstrated that the Si content and Si/Al ratio on the surfaces of A(5)FS, A(10)FS, and A(15)FS catalysts gradually increase with the AHFS concentration. This increase could be attributed to the formation of a silica layer on the catalyst's outer surface. Fig. 3(d), (e), and (f) indicated a decrease in the Si/Al ratio on the surface of A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B catalysts compared to that of the A(10)FS catalyst. Moreover, a decrease in the content of Ni, Fe, and V on the surface of all activated catalysts was observed. The reduction of these metallic contaminants implied that a portion of the pore structure on the catalyst surface was restored, thereby facilitating the improved utilization of the active sites within the catalysts.

To analyze the concentrations and types of acid sites on the catalysts, pyridine-infrared spectroscopy was employed, and the results of the analytical data are listed in Table 4. As shown in Table 4, the A(5)FS catalyst exhibited a lower concentration of Lewis acid sites compared to the S-FCC catalyst, while the concentration of Brønsted acid sites was higher. Previous research has shown that the amounts of framework Al and extra-framework Al in the catalyst were related to the concentration of Brønsted and Lewis acid sites, respectively [33]. The acid treatment reduced the content of extra-framework Al in the catalyst, leading to a decrease in the concentration of Lewis acid sites in the A(5)FS catalyst. Moreover, the treatment removed metallic contaminants from the catalyst surface, thereby exposing more framework Al and increasing the concentration of Brønsted acid sites. With an increase in the concentration of AHFS, the breaking extent of the Si–O–Al bond in the framework also increased. As a result, more framework Al was replaced by Si, which decreased the concentration of Brønsted acid sites. However, during the migration process, some of the replaced Al remained within the catalyst and transformed into extra-framework Al, thereby increasing the concentration of the Lewis acid sites. Consequently, compared to A(5)FS, A(10)FS exhibited an increased concentration of the Lewis acid sites, while the concentration of Brønsted acid sites decreased. In the case of A(15)FS, the amounts of framework Al was removed, resulting in the formation of a silica layer that covered the acid sites on the external surface of the catalyst. As a result, both the concentrations of Lewis and Brønsted acid sites decreased [34,35].

When the A(10)FS catalyst underwent treatment with NaOH solutions with different concentrations, the Si–O–Al bonds in the catalyst framework were cleaved, releasing adjacent Al atoms and converting the initial Brønsted acid sites into Lewis acid sites [36,37]. In addition, the NaOH solution removed the silica layer that had been deposited on the

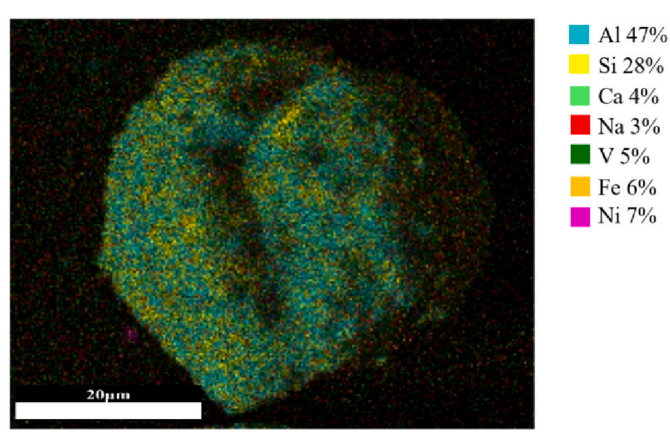

Fig. 2. SEM-EDS maps of the content of elements on S-FCC catalyst.

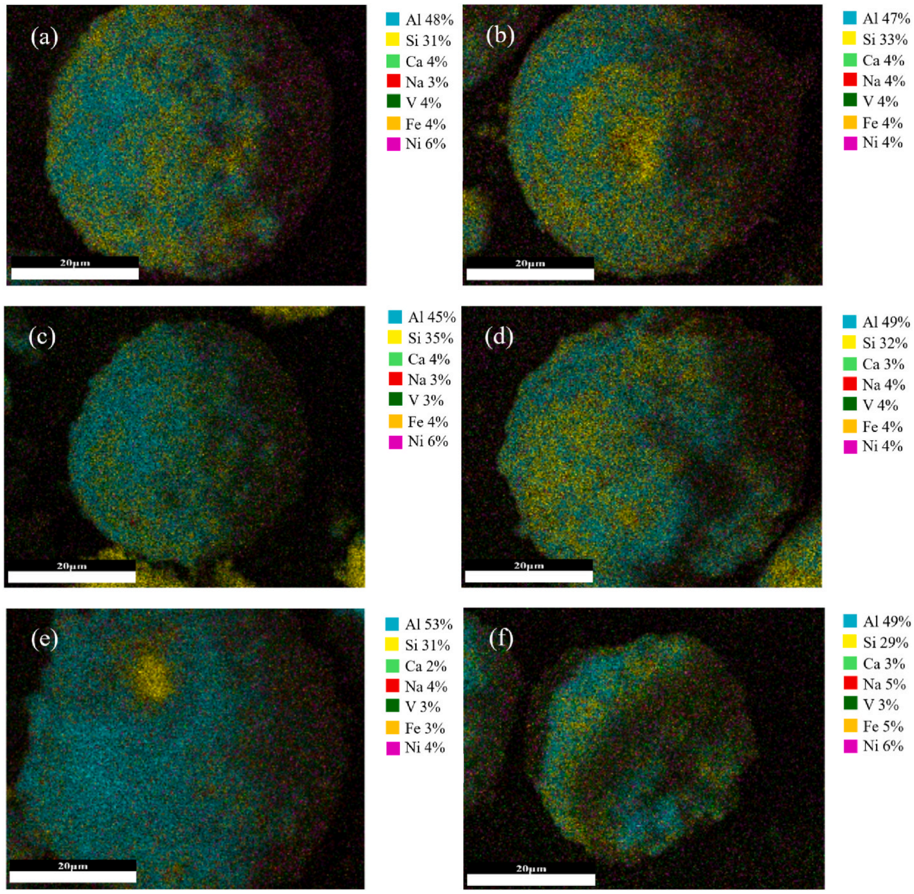

Fig. 3. SEM-EDS maps of the content of elements on activated catalysts. (a) A(5)FS, (b) A(10)FS, (c) A(15)FS, (d) A(10)FS-0.1B, (e) A(10)FS-0.3B and (f) A(10)FS-0.5B.

Table 4
Acidic properties of spent FCC catalyst and activated catalysts.

Sample	Lewis acid site (μmol/g)	Brønsted acid site (μmol/g)	Total acid site (μmol/g)	Lewis acid site/ Brønsted acid site
S-FCC	34.63	11.16	45.79	3.10
A(5)FS	30.90	12.03	42.93	2.57
A(10)FS	34.44	8.40	42.84	4.10
A(15)FS	32.91	5.38	38.29	6.12
A(10)FS-0.1B	34.14	9.82	43.96	3.48
A(10)FS-0.3B	45.90	12.14	58.04	3.78
A(10)FS-0.5B	36.38	10.96	47.34	3.32

external surface of the catalyst, thereby re-exposing the acidic sites of the catalyst. Consequently, the alkali-treated catalyst exhibited a higher concentration of acid sites than the A(10)FS catalyst. The concentrations of total acid sites in the catalysts increased with increasing alkali concentration, followed by decreases. Among them, the A(10)FS-0.3B catalyst exhibited the highest concentration of total acid sites. This finding indicated that an appropriate alkali concentration can enhance the acid site concentration of the catalyst. However, excessively high concentrations resulted in the collapse of the catalyst structure, leading to the destruction of acid sites and lower concentrations of Lewis and Brønsted acid sites.

3.2. *Product distribution from catalytic pyrolysis of DH-MMPW*

In this work, the catalytic pyrolysis of DH-MMPW was investigated using spent FCC catalyst and activated catalysts in a confined fluidized bed reactor. The distribution of catalytic pyrolysis products is shown in Fig. 4. The mass balance for all experiments conducted ranged from 91.93 wt% to 97.54 wt%. The slight loss in mass balance was attributed

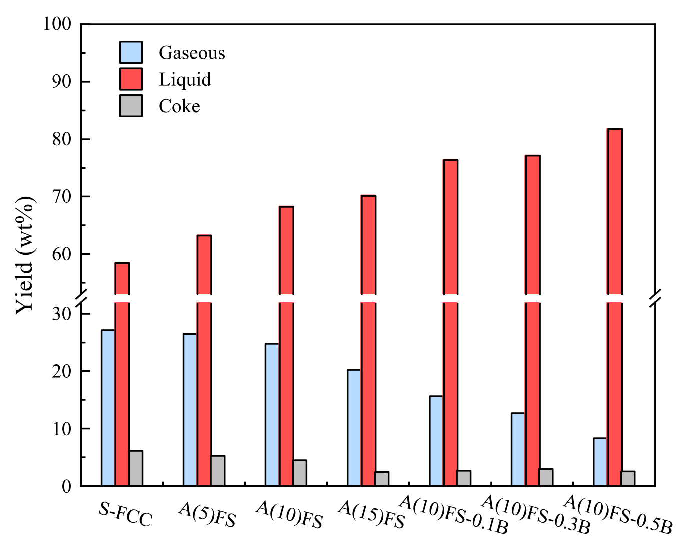

Fig. 4. Product distribution of catalytic pyrolysis over spent FCC catalyst and activated catalysts.

to the adherence of a small amount of feedstock and product to the inner wall of the inlet piping, reactor, and outlet piping due to the high viscosity of the feedstock and the long length of the experimental setup. Consequently, some feedstock and products were inevitably lost.

The catalytic pyrolysis of DH-MMPW using spent FCC catalyst and the activated catalyst primarily produced liquid products. This was attributed to the short residence time within the confined fluidized bed reactor, which facilitated the formation of liquid hydrocarbon products. According to the results in Fig. 4, the S-FCC catalyst exhibited the lowest yield of 58.50 wt% for the liquid product while producing the highest yields of 27.13 wt% and 6.12 wt% for the gaseous and coke, respectively. From the textural properties of the catalyst in Tables 3 and it is evident that the S-FCC catalyst had a relatively high proportion of micropores and the highest specific surface area, resulting in a higher yield of gaseous product. Moreover, the higher specific surface area facilitated a greater number of acidic sites for the reactants to undergo the cracking reaction. However, the presence of metallic contaminants on the surface of the spent FCC catalyst caused pore blockage and affected pore connectivity. Consequently, some intermediates became trapped within the catalyst's pores, leading to the generation of coke precursors or smaller molecules. These coke precursors were challenging to diffuse outside of the pores, which increased the yield of coke and deactivated the catalyst.

Compared to S-FCC, the A(5)FS, A(10)FS, and A(15)FS catalysts exhibited higher liquid yields and lower gas and coke yields due to their improved structural properties. The increased external surface area of the catalyst facilitated the initial cracking reaction of polyolefin macromolecules, while the improved pore connectivity of the catalyst and increased mesopore and macropore volume promoted the diffusion of cracking products. Furthermore, the decreased concentration of acid site led to a decrease in the depth of the cracking reaction. All these factors contributed to higher yields of liquid products. As the concentration of AHFS increased, the yield of the liquid products increased while the yield of the gaseous and coke gradually decreased. The A(5)FS and A(10)FS catalysts had slightly lower specific surface area, micropore volume, and total acid site concentration compared to the S-FCC catalyst. As a result, the yields of the gaseous products were reduced by 2.43% and 8.74%, respectively. However, the A(5)FS and A(10)FS catalysts showed an increase in external surface area and meso-macropore volume, leading to an 8.15% and 16.74% increase in liquid product yield, respectively. Although the A(15)FS had poorer textural properties compared to the A(10)FS, it achieved significantly higher yields of liquid products with a remarkable difference of 28.03% compared to S-FCC. Additionally, there was a relative decrease of 25.47% in gaseous product yields. The lower concentration of acid site and specific surface area under the same reaction conditions resulted in lower catalytic activity and an increased tendency of thermal cracking reactions. However, the thermal cracking reaction favored the generation of liquid products [9], resulting in a higher yield of liquid products for A(15)FS.

To investigate the impact of removing the silica layer on the catalyst, A(10)FS was treated with different concentrations of NaOH solutions. Compared to A(10)FS, the gaseous product yields of A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B decreased by 36.87%, 48.79%, and 66.44%, respectively. Conversely, the liquid product yields increased by 10.60%, 11.62%, and 17.70%, respectively. These results indicated that as the alkali concentration increased, the catalyst's specific surface area and pore volume gradually decreased, thereby shortening the reaction path. As a result, the residence time of reactants and products in the catalyst's pores decreased, leading to the formation of hydrocarbon compounds with longer carbon chains.

3.3. Gaseous products from catalytic pyrolysis of DH-MMPW

The selectivity for the components (H_2, C1–C4) in the gaseous products is shown in Fig. 5. The results indicated that the gaseous products primarily consisted of C3 and C4 hydrocarbons, which accounted for 82.15–92.84% of the gaseous products. With the S-FCC catalyst, the feedstock underwent deeper cracking and hydrogen transfer reactions on Brønsted acid sites, resulting in a higher selectivity of alkanes in the gas products compared to the activated catalysts. Consequently, the S-FCC catalyst exhibited the lowest selectivity for light olefins ($C_2^=$-$C_4^=$) at 57.02%. The selectivity of the components in the gaseous products of the A(5)FS, A(10)FS, and A(15)FS catalysts significantly differed from those of the S-FCC catalyst, particularly for propylene, butane, and butene. As the concentration of Brønsted acid sites decreased, the selectivity of butane significantly decreased, while the opposite occurred for propene and butene. Compared to S-FCC, the selectivity for light olefins in the gaseous products increased by 5.79%, 18.25%, and 19.49%, respectively. For the A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B catalysts, the selectivities for light olefins in the gaseous products were significantly increased by 36.27%, 38.19%, and 54.25%, respectively, compared to S-FCC catalyst. This increase can be attributed to the reduction in specific surface area and pore volume, as well as the increased pore sizes. These factors hindered the intermolecular hydrogen transfer reactions, resulting in an increased olefin selectivity. These findings demonstrated that activation of the S-FCC catalyst effectively suppressed secondary reactions during catalytic pyrolysis, reducing gas production yield and promoting the production of high-value light olefins with high selectivity. This contributed to the utilization of the waste plastics.

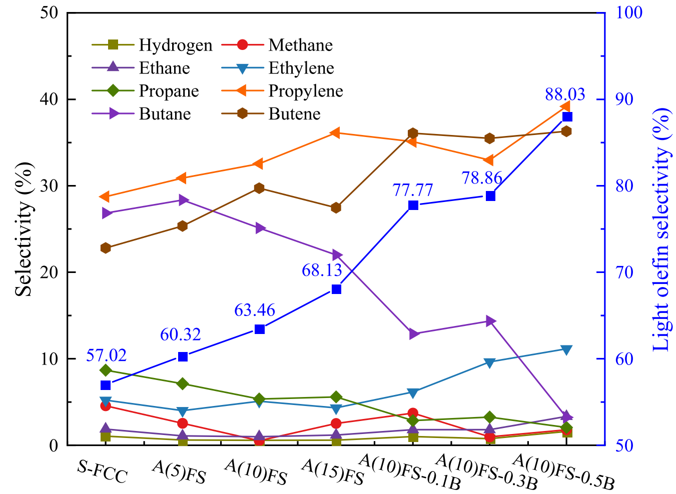

Fig. 5. Selectivities of the components in the gaseous products over spent FCC catalyst and activated catalysts.

3.4. Liquid products from catalytic pyrolysis of DH-MMPW

To evaluate the impact of the activated catalyst on the yield of light fuels in the liquid products, the fraction distribution of the liquid products was analyzed using simulated distillation. The results, shown in Fig. 6, indicated that the S-FCC catalyst yielded the highest gasoline fraction, accounting for 34.05 wt% of the total product yield. However, the activated catalysts resulted in varying degrees of reduction in gasoline fractions. A(10)FS exhibited the smallest reduction at only 3.96%, while A(10)FS-0.5B showed the most significant reduction at 45.93%. The liquid products generated by the activated catalysts consisted predominantly of the diesel fraction, accounting for 32.98–45.66 wt% of the total product yield. Compared to the S-FCC catalyst, the diesel fraction yields of A(5)FS, A(10)FS, and A(15)FS increased by 46.73%–63.56%, while the diesel fraction yields of A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B were 0.95–1.19 times higher. The trend for the heavy oil fraction in the liquid product was similar to that of the diesel fraction. The S-FCC catalyst exhibited the lowest heavy oil yield, while the heavy oil yields of A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B significantly increased with increasing alkali concentrations. Given the same reaction

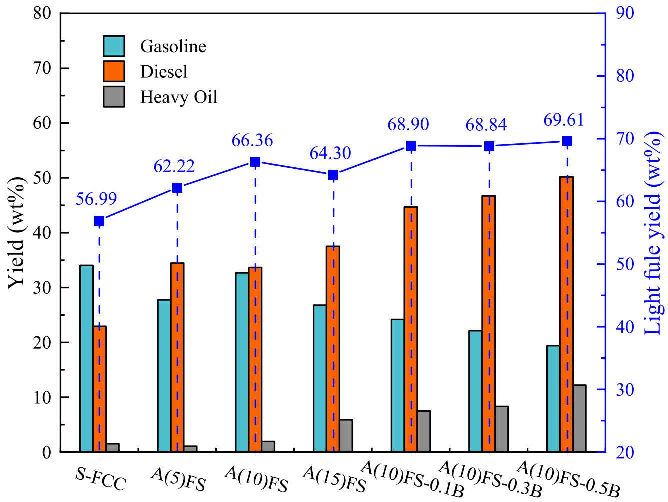

Fig. 6. Liquid product analysis over spent FCC catalyst and activated catalysts.

conditions, the different fraction distributions in the liquid products were mainly attributed to the acid-site concentrations and pore structures of the catalysts. The S-FCC catalyst possessed abundant pore structures and high acid-site concentrations than the activated catalysts. This allowed the reactants and reaction intermediates to undergo longer reaction paths, resulting in deeper cracking on the acidic sites within the pores and promoting the efficient cracking of waste plastics into hydrocarbon products with short carbon chains. As a result, there was a lower yield of heavy oil and a higher yield of gasoline in the liquid product. The A(5)FS, A(10)FS, and A(15)FS catalysts improved the initial cracking reactions of waste plastics and reduced the diffusion limitations of reactants and products within the catalyst pores by increasing the external surface area and meso-macropore volume [38]. However, the lower concentration of acid sites hindered the further conversion of the primary products, leading to a significant increase in diesel yield compared to the S-FCC catalyst. For the A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B catalysts, the destruction of the pores resulted in lower catalytic activity, thereby significantly increasing the yields of diesel and heavy oil in the liquid products.

Based on the simulated distillation results, it is evident that although the S-FCC catalyst achieved the highest gasoline yield, a significant amount of feedstock was converted into gaseous and coke products, resulting in a light fuel yield of only 56.99 wt% of the total yield. However, the activated catalysts exhibited a considerable improvement in light fuel yields compared to the S-FCC catalyst. The A(5)FS and A(10)FS catalysts increased the light fuel yields by 9.18% and 16.44%, respectively, while reducing the gaseous product yields by 2.43% and 8.74%, respectively, and showing a slight increase in heavy oil production. This indicated that the A(5)FS and A(10)FS catalysts efficiently converted the feedstock, with higher selectivity for light fuel. The A(15)FS catalyst showed a lower yield of the light fuel and a higher yield of the heavy oil than A(10)FS due to the lower acid site concentration and specific surface area. The A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B catalysts exhibited higher selectivities for both light fuel and heavy oil. This was mainly achieved by adjusting the pore structures of the catalysts. However, the lower catalytic activities shifted the liquid product fraction towards heavier components, resulting in higher diesel and heavy oil yields.

3.5. Effect of activated catalysts on the catalytic pyrolysis of H-MMPW

The activated catalyst showed a significantly higher yield of light fuel, which may be related to the unconverted straight-run diesel in the feedstock. To investigate the impact of the activated catalysts on the catalytic pyrolysis of H-MMPW, catalytic pyrolysis of straight-run diesel was conducted under identical reaction conditions. The results are shown in Figs. S1–3. Due to the smaller molecular size of straight-run diesel compared to the waste plastics, there was less diffusion resistance and shorter residence times in the catalysts with a high percentage of meso-macroporous structures. As a result, compared to DH-MMPW, the yield of gaseous product was lower, while the yield of liquid products was higher. Additionally, the yield of gasoline was lower, while the yield of diesel was higher. Theoretical calculations were also performed to determine the product yields (light olefins, gasoline, and diesel) for catalytic pyrolysis of H-MMPW according to Eq. (2). As shown in Fig. 7, S-FCC yielded the highest light olefin and gasoline,18.77 wt% and 36.45 wt%, respectively. However, the diesel yield of S-FCC was only 7.57 wt %, indicating that the diesel in the liquid product from catalytic pyrolysis of DH-MMPW using S-FCC was mainly derived from the diesel product formed by catalytic pyrolysis of the straight-run diesel. A(5)FS, A(10)FS, and A(15)FS produced lower yields of light olefins and gasoline than S-FCC but higher diesel yields. Among these catalysts, A(10)FS yielded the highest gasoline and diesel, 35.20 wt% and 23.49 wt%, respectively, while A(15)FS yielded the lowest gasoline and diesel yields, 30.40 wt% and 20.58 wt%, respectively. The catalyst activities were lower for A(10)FS-0.1B, A(10)FS-0.3B, and A(10)FS-0.5B due to their lower specific surface areas and pore volumes. As a result, the yields of light olefins and gasoline were lower, while the yield of diesel was higher. Additionally, both gasoline and diesel yields gradually decreased with increasing alkali concentration. Regarding light fuel, although S-FCC produced the highest gasoline yield, the lowest light fuel yield was only 44.02 wt%. The activated catalysts showed higher light fuel yields, 14.06–33.33%, than S-FCC. Among them, A(10)FS exhibited the highest light fuel yield of 58.69 wt% with a high proportion of gasoline, while A(10)FS-0.1B exhibited a slightly lower light fuel yield of 57.28 wt% with a high proportion of diesel. These findings demonstrated that activation of the spent FCC catalyst enhanced the production of light fuel.

In addition, Fig. 8 illustrates the effects of the spent FCC catalyst and activated catalysts on H-MMPW conversion. Since the product obtained from catalytic pyrolysis of straight-run diesel did not contain heavy oil, the heavy oil in the product from DH-MMPW was derived from the heavy oil generated by the conversion of H-MMPW. Assuming that the heavy oil in the liquid product was unconverted feedstock, the conversion of H-MMPW over the S-FCC catalyst and the activated catalysts was calculated. The results showed that the conversions of H-MMPW over S-FCC, A(5)FS, and A(10)FS were greater than 96 wt%, which indicated high activities. Among these catalysts, A(5)FS exhibited the highest H-

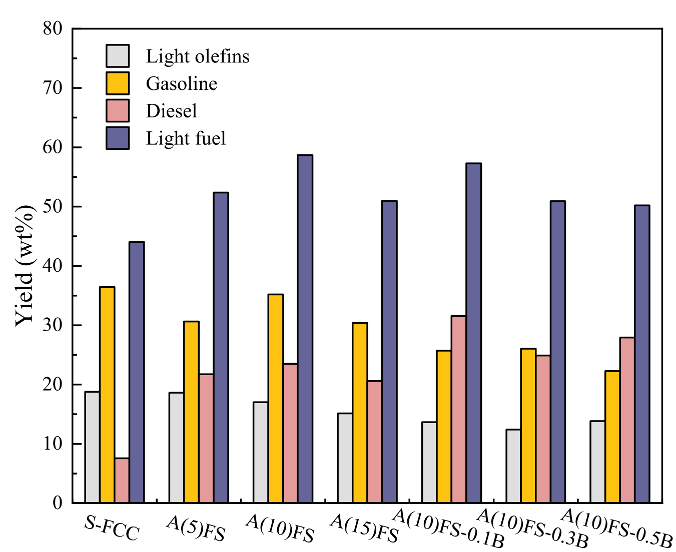

Fig. 7. Product yields from catalytic pyrolysis of H-MMPW over the spent FCC catalyst and activated catalysts.

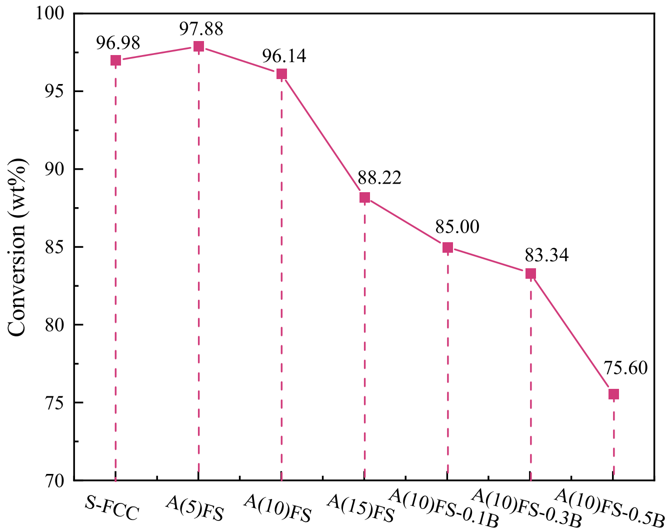

Fig. 8. Conversion rates of H-MMPW over the spent FCC catalyst and activated catalysts.

MMPW conversion of 97.88 wt%. Although the conversion of H-MMPW over the A(10)FS catalyst was 1.78% lower, it resulted in a higher light fuel yield by 12.05% and gasoline yield by 14.88%. The A(15)FS, A(10)FS-0.1B, and A(10)FS-0.3B catalysts exhibited moderate catalytic performance, with H-MMPW conversions of 88.22 wt% 85.00 wt%, and 83.34 wt%, respectively. The lower catalytic activity of the A(10)FS-0.5B catalyst, attributed to its lower specific surface area and pore volume compared to A(10)FS, resulted in the conversion rate of H-MMPW below 80 wt%. Consequently, A(10)FS-0.5B exhibited the lowest light fuel yield of 50.21 wt% among the activated catalysts. This indicated that the impact of catalyst activity on H-MMPW conversion should be considered in conjunction with the enhanced in light fuel yield. A high catalyst activity would reduce the light fuel yield. Conversely, it inhibited the conversion of the feedstock, leading to a higher in the heavy oil fraction in the liquid product.

4. Conclusion

The development of chemical recycling technologies for waste plastics plays a crucial role in solid waste treatment and the advancement of the green economy. Catalytic pyrolysis with spent FCC catalysts is an efficient method for recycling municipal mixed plastic waste, and it reduces the cost of the catalyst and converts the waste plastics into valuable light fuel. In this study, the spent FCC catalyst was activated via sequential treatment with nitric acid and AHFS to improve its textural properties and enhance the yield of light fuel. The activation process increased external surface area and meso-macroporous volume of the catalyst, which facilitated initial cracking reaction of polyolefin macromolecules on the catalyst's surface and improved the diffusion rates of reactants and products within the pores. Consequently, the light fuel yields of the A(5)FS and A(10)FS catalysts were 21.67% and 29.76% higher than that of the spent FCC catalyst, with yields of 62.22 wt% and 66.36 wt%, respectively, and the H-MMPW conversions were all greater than 96 wt%. Similarly, compared to the spent FCC catalyst, the light fuel yield of the A(15)FS catalyst was higher, although the H-MMPW conversion was lower. Further treatment of the A(10)FS catalyst with NaOH solutions reduced its specific surface area and pore volume. Consequently, the H-MMPW conversion decreased, while the yield of light fuel increased due to the presence of unconverted straight-run diesel. Among the catalysts, A(10)FS-0.1B showed the highest light fuel yield of 68.90 wt%, but the proportion of diesel in the light fuel was also significantly increased. These findings indicated that the activation of the spent FCC catalyst increased the yield of light fuels. Additionally, the gaseous product contained a significant proportion of light olefin. This approach is promising for the high-value utilization of waste plastics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Key R&D Program of China [grant numbers 2019YFC1906305].

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.joei.2024.101556.

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